

QUESTION BANK
CHAPTER 12-ATOMS

SECTION A

Multiple Choice Questions (1 Mark each)

Q.1 In Bohr's theory of model of hydrogen atom, name the physical quantity which equals to an integral multiple of $h/2\pi$?

- (a) Momentum (b) Angular momentum
(c) Angular frequency (d) Angular velocity

Q.2 What is the relation between 'n' and radius 'r' of the orbit of electron in hydrogen atom according to Bohr's theory?

- (a) 'r' is directly proportional to n (b) 'r' is directly proportional to n^2
(c) ' n^2 ' is directly proportional to r (d) 'r' is inversely proportional to n^2

Q.3 What is Bohr's frequency condition?

- (a) $h\nu = E_i - E_f$ (b) $h\nu = E_f - E_i$ (c) $h\nu = E - E_f$ (d) $h = E_i - E_f$

Q.4 Write the expression for Bohr's radius in hydrogen atom?

- (a) $r = 4\pi\epsilon_0 \times h^2 / 4\pi^2 m e^2$ (b) $r = 4\pi\epsilon_0 \times h^2 / 4\pi^2 m e^2$
(c) $r = 4\pi\epsilon_0 \times h^2 / 4\pi m e^2$ (d) $r = 4\pi\epsilon_0 \times h / 4\pi^2 m e^2$

Q.5 Name the spectral series of hydrogen atom lying in visible region?

- (a) Paschen series (b) Pfund series (c) Brackett series (d) Balmer series

Q.6 The total energy of an electron in 1st excited state of hydrogen atom is about -3.4 eV.

What is the kinetic energy of electron in this state?

- (a) -3.4 eV (b) 3.4 eV (c) 0.34 eV (d) -0.34 eV

Q.7. Which of the following spectral series in hydrogen atom gives spectral line of 4860 Å?

- (a) Lyman (b) Balmer (c) Paschen (d) Brackett

Q.8 When hydrogen atom is in first excited level, its radius is

- (a) same (b) half (c) twice (d) 4 times

Q.9 Rutherford model of atom was unstable because

- (a) nuclei will break down (b) electron move in circular orbit
(c) orbiting electrons radiate energy (d) electrons are repelled by the nucleus

Assertion –Reason type questions(1 mark each)

Answer: A Both are correct and reason is correct explanation of assertion.

Answer: B Both are correct but reason is not the correct explanation of assertion.

Answer: C Reason is wrong.

Answer: D Both are wrong.

Q10. Assertion: According to Bohr's atomic model the ratio of angular momenta of an electron in first excited state and in ground state is 2:1.

Reason: In a Bohr's atom the angular momentum of the electron is directly proportional to the principal quantum number.

Q11. Assertion: The positively charged nucleus of an atom has a radius of almost 10^{-15}m .

Reason: In α -particle scattering experiment, the distance of closest approach for particles is $\approx 10^{-15}\text{m}$.

Q12. Assertion: For the scattering of α -particles at a large angles, only the nucleus of the atom is responsible.

Reason: Nucleus is very heavy in comparison to α particle.

Q13. Assertion: Bohr had postulated that the electrons in stationary orbits around the nucleus do not radiate

Reason: According to classical physics all moving electrons radiate.

Q14. Assertion: Atoms are not electrically neutral.

Reason: Number of protons and electrons are different.

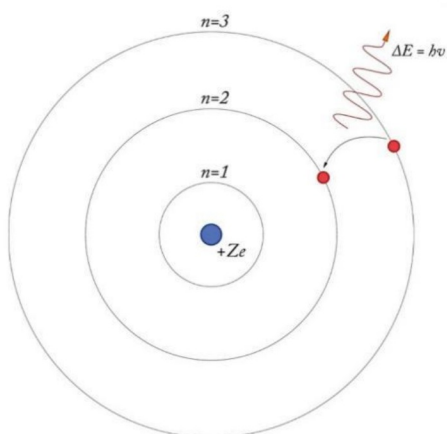
Case study based questions (4 marks)

Q 15. BOHR ATOM MODEL

Rutherford gives Rutherford's model of the atom after performing an Alpha particle scattering experiment.

This model is a modification of the earlier Rutherford Model. According to this model, an atom consists of a small, positively-charged nucleus and negatively-charged electrons orbiting around it in an orbital. These orbital can have different sizes, energy, etc. And the energy of the orbit is also related to its size. The lowest energy is found in the smallest orbit. So if the electron is orbiting in n th orbit then we will study about its Velocity in n th orbital, Radius of n th orbital, Energy of

electron in n th orbit, etc. Energy is also emitted due to the transition of electrons from one orbit to another orbit. This energy is emitted in the form of photons with different wavelengths. This wavelength is given by the Rydberg formula. When electrons make transitions between two energy levels in an atom various spectral lines are obtained. The emission spectrum of the hydrogen atom has been divided into various spectral series like Lyman series, Balmer series, Paschen series etc.



Bohr model

(I). The formula which gives the wavelength of emitted photon when electron jumps from higher energy state to lower was given by

- (a) Balmer (b) Paschen (c) Lyman (d) Rydberg

(II). What is true about Bohr's atomic Model?

- (a) His model was unique totally different from other
 (b) His model is a modification of Rutherford atomic model.
 (c) His model is a modification of Thomson atomic model.
 (d) None of the above

(III). Bohr's atomic model is applicable for

- (a) All types of atoms (b) Only for hydrogen atom
 (c) For hydrogen like atoms (d) For H₂ gas.

(IV). The cause of rejection of Rutherford atomic model was

- (a) It was totally wrong (b) It could not justify its stability
 (c) Rutherford was unable to explain it (d) none of the above.

Short Answer type questions (2 marks each)

- Q16. The ground state energy of hydrogen atom is -13.6eV . What are the kinetic energies and potential energies of the electron in the ground state and second excited state.
- Q17. Define impact parameter. Represent diagrammatically the shape of trajectory of alpha particles.
- Q18. Show graphically the variation of radius of orbit with principal quantum number n .
- Q19. The radius of innermost electron orbit of a hydrogen atom is $5.3 \times 10^{-11}\text{ m}$. What is the radius of orbit in the second excited state?
- Q20. Define ionization energy. How would the ionization energy change when electron in hydrogen atom is replaced by the particle of mass 200 times that of the electron but having the same charge? [Given Rydberg constant, $R = 10^7\text{ m}^{-1}$] (2016)
- Q21. Write two important limitations of Rutherford nuclear model of the atom. (2017)
- Q22. Calculate the de-Broglie wavelength of the electron orbiting in the $n = 2$ state of hydrogen atom. (2016)
- Q23. Derive an expression for the radius of n th Bohr's orbit in Hydrogen atom.

Short answer type questions. (3 marks each)

- Q24. In a Geiger-Marsden experiment, calculate the distance of the closest approach to the nucleus of $Z = 80$, when an α -particle of 8 MeV energy impinges on it before it comes momentarily to rest and reverses its direction. How will the distance of the closest approach be affected when the kinetic energy of the α -particle is doubled?
- Q25. Derive an expression for the frequency of radiation emitted when a hydrogen atom de-excites from level n to level $(n - 1)$. Also show that for large values of n , this frequency equals to classical frequency of revolution of an electron. (CBSE SQP 2020-21)

Long answer type questions. (5 marks each)

- Q26. (i) Using Bohr's postulates, derive an expression for the total energy of the electron in the stationary states of the hydrogen atom. (CBSE 2013)
- (ii) Using Rydberg formula, calculate the wavelengths of the spectral lines of the first member of the Lyman series and of the Balmer Series.

ANSWERS (SECTION A)

MCQs - Answers

- 1.(b) Angular momentum
2. (b) 'r' is directly proportional to n^2
- 3.(b) $h\nu = E_f - E_i$
4. (b) $r = 4\pi\epsilon_0 \times \frac{h^2}{4\pi^2 m e^2}$
5. (d) Balmer series
6. (b) $K.E. = -E = -(-3.4) = 3.4 \text{ eV}$
7. (b) Balmer
8. (d) 4 times
9. (c) orbiting electrons radiate energy

Assertion and Reason- Answers

10. Correct answer: A
11. Correct Answer: A
12. Correct Answer: A
13. Correct Answer: B
14. correct Answer D

Case Study Based question- Answers

15. (I) (d) Rydberg
- (II) (b) His model is a modification of Rutherford atomic model.
- (III) (c) For hydrogen like atoms
- (IV) (b) It could not justify its stability

Short answer type questions (2M)- Answers

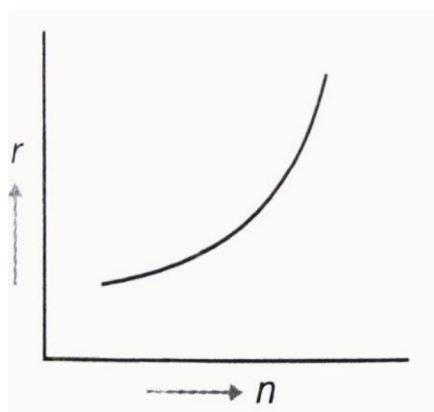
16. $K.E. = 13.6 \text{ eV}$, $P.E. = -27.2 \text{ eV}$ in the ground state
 $K.E. = 1.51 \text{ eV}$, $P.E. = -3.02 \text{ eV}$ in the second excited state

17. Impact parameter perpendicular distance of the velocity vector of a-particle from the central line of the nucleus of the atom is called impact parameter (b).

for figure refer page no 115

$$b = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2 \cot \theta/2}{K}$$

18. Orbital radius is directly proportional to the square of principal quantum number



19. For $n=1$, $r_1 = 5.3 \times 10^{-11} \text{ m}$ for ground state therefore $r_2 = 4 r_1$, and $r_3 = 9 r_1$

$r_2 = 2.12 \times 10^{-10} \text{ m}$ $r_3 = 4.77 \times 10^{-10} \text{ m}$

$\lambda_{\min} = 3.646 \times 10^{-7} \text{ m} = 364.6 \text{ nm}$. **This wavelength lies in the ultraviolet region**

20. Ionization energy: It is the minimum amount of energy required to remove an electron from the outermost orbit of an atom in its ground state.

$$E_0 = \frac{me^4}{8\epsilon_0^2 h^2} \text{ i.e., } E_0 \propto m$$

Therefore, ionization energy will become 200 times.

21. According to Rutherford's atomic model, the electron orbiting around the nucleus continuously radiates energy because of its acceleration due to which the atom will not remain stable. So, Rutherford could not explain the stability of an atom.

(ii) Since electron spirals inwards, its angular velocity and frequency change continuously, therefore it will emit a continuous spectrum. Therefore, this model could not explain the line spectra of hydrogen.

22. Kinetic Energy is given by the formula

$$E = \frac{13.6 \text{ eV}}{n^2} = \frac{13.6 \text{ eV}}{2^2} = \frac{13.6 \text{ eV}}{4}$$

$$= 3.4 \times 1.6 \times 10^{-19} J$$

De Broglie's wavelength,

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} E}} = 0.067 \text{ nm}$$

23. from Bohr's first postulate,

coloumb force F = centripetal force F_{cp}

$$\therefore \frac{Ze^2}{4\pi\epsilon_0 r_n^2} = \frac{m_e v_n^2}{r_n}$$

$$v_n^2 = \frac{Ze^2}{4\pi\epsilon_0 r_n m_e} \dots\dots\dots(1)$$

according to Bohr's second postulate,

$$m_e r_n v_n = \frac{nh}{2\pi}$$

$$m_e^2 r_n^2 v_n^2 = \frac{n^2 h^2}{4\pi^2}$$

$$v_n^2 = \frac{n^2 h^2}{4\pi^2 m_e^2 r_n^2} \dots\dots\dots(2)$$

iv from equation (1) and (2)

$$\frac{n^2 h^2}{4\pi^2 m_e^2 r_n^2} = \frac{Ze^2}{4\pi\epsilon_0 r_n m_e}$$

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m_e Z e^2}$$

$$r_n = \left(\frac{\epsilon_0 h^2}{\pi m_e^2 Z e^2} n^2 \right)$$

this is the required equation for radius of n^{th} orbit.

Short answer type questions (3M)- Answers

24. At closest approach, all K.E. of α -particles is completely converted into the P.E. of α -particle.

$$K.E. = 8MeV = 8 \times 10^6 \times 10^{-19} J.$$

K.E. is converted into P.E.

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{Ze2e}{r_0} = 12.8 \times 10^{-13}$$

$$9 \times 10^9 \times \frac{80 \times 2 \times 16 \times 10^{-19}}{r_0} = 12.8 \times 10^{-13}$$

$$r_0 = \frac{9 \times 10^9 \times 160 \times (1.6 \times 10^{-19})^2}{12.8 \times 10^{-13}} (m)$$

$$r_0 = \frac{9 \times 1.6 \times 1.6 \times 1.6 \times 10^{-14}}{12.8}$$

$$= 2.88 \times 10^{-14} m.$$

As we see that $K.E. = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2e}{r_0}$

Hence, $K.E. \propto \frac{1}{r_0}$

If K.E. become twice, closest approach will be halved.

Hence, $r'_0 = \frac{r_0}{2}$

25. from Bohr's theory frequency of the radiation emitted when an electron de- excites from level n_2

to level n_1 is given as

$$f = \frac{2\pi^2 mk^2 z^2 e^4}{h^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Given $n_1 = n-1$, $n_2 = n$

$$f = \frac{2\pi^2 mk^2 z^2 e^4}{h^3} \frac{2n-1}{(n-1)^2 n^2}$$

For large n , $2n-1=2n$, $n-1=n$ and $z=1$

Thus, $f = \frac{4\pi^2 mk^2 z^2 e^4}{n^3 h^3}$

Which is same as orbital frequency of electron in n^{th} orbit

$$f = \frac{v}{2\pi r} = \frac{4\pi^2 mk^2 z^2 e^4}{n^3 h^3}$$

Long answer question (5M) - Answers

26. $mvr = \frac{nh}{2\pi}$

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$r = \frac{e^2}{4\pi\epsilon_0 mv^2}$$

$$r = \frac{\epsilon_0 n^2 h^2}{\pi m e^2}$$

$$\begin{aligned}\text{P.E. } U &= -\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r} \\ &= \frac{me^4}{4\epsilon_0 n^2 h^2}\end{aligned}$$

$$\begin{aligned}\text{K.E.} &= \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{nh}{2\pi m r} \right)^2 \\ &= \frac{n^2 h^2 \pi^2 m^2 e^4}{8\pi^2 m \epsilon_0^2 n^4 h^4}\end{aligned}$$

$$\text{K.E.} = \frac{me^4}{8\epsilon_0^2 n^2 h^2}$$

$$\text{T.E.} = \text{K.E.} + \text{P.E.}$$

$$= -\frac{me^4}{8\epsilon_0^2 n^2 h^2}$$

(b) Rydberg formula for first member of Lyman series

$$\begin{aligned}\frac{1}{\lambda} &= R \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \\ \lambda &= \frac{4}{3R}\end{aligned}$$

For first member of Balmer Series

$$\begin{aligned}\frac{1}{\lambda} &= R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \\ \lambda &= \frac{36}{5R}\end{aligned}$$

SECTION B

Multiple choice questions (1 mark each)

Q.1 Why did Thomson's atomic model fail?

- (a) Thomson's model failed to explain the scattering of alpha particles through large angles in Rutherford experiment.
- (b) Thomson's model failed to explain the scattering of alpha particles through small angles in Rutherford experiment.
- (c) Thomson's model failed to explain the scattering of beta particles through large angles in Rutherford experiment.
- (d)) Thomson's model failed to explain the scattering of beta particles through small angles in Rutherford experiment.

Q.2 How much is the radius of Bohr's inner most orbit?

- (a) $r=0.51 \text{ }^{\circ}\text{A}$ (b) $r=0.35 \text{ }^{\circ}\text{A}$ (c) $r=0.23 \text{ }^{\circ}\text{A}$ (d) $r=0.53 \text{ }^{\circ}\text{A}$

Q3. The transition from the state $n = 5$ to $n = 1$ in a hydrogen atom results in UV radiation.

Infrared radiation will be obtained in the transition

- (a) $2 \rightarrow 1$ (b) $3 \rightarrow 2$ (c) $4 \rightarrow 3$ (d) $6 \rightarrow 2$

Q.4 The energy of hydrogen atom in its ground state is -13.6eV . The energy of level corresponding to $n=5$ is

- (a) -0.54eV (b) -5.40eV (c) -0.85eV (d) -2.75eV

Q.5 Hydrogen atom are excited from ground state of the principal quantum number 4 then number of spectral lines observed will be

- (a) 3 (b) 6 (c) 5 (d) 2

Q.6 In Bohr model of hydrogen atom which of the following is quantised?

- (a) linear velocity of electron (b) angular velocity of electron
(c) linear momentum (d) angular momentum

Assertion –Reason Type Questions (1 mark)

Answer: A Both are correct and reason is correct explanation of assertion.

Answer: B Both are correct but reason is not the correct explanation of assertion.

Answer: C Reason is wrong.

Answer: D Both are wrong.

Q7. Assertion: The force of repulsion between atomic nucleus and α -particle varies with distance according to inverse square law.

Reason: Rutherford did α -particle scattering experiment

Q8. Assertion : Total energy of revolving electron in any stationary orbit is negative.

Reason : Energy is a scalar quantity. It can have positive or negative value.

Q9.Assertion:Balmer series lies in the visible region of electromagnetic spectrum

Reason: $1/\lambda=R(1/2^2-1/n^2)$, where $n=3,4,5,\dots$

Case study based question (5 marks)

Q 10. HYDROGEN EMISSION SPECTRA

Hydrogen spectrum consist of discrete bright lines in dark background is known as Hydrogen Emission spectrum. There is one more type of Hydrogen spectrum that exist where we get dark lines on bright background. It is known as absorption spectra.

Balmer found empirical formula by the observation of a small part of the spectrum and it is represented by $1/\lambda = R(1/2^2-1/n^2)$ where $n = 3,4,5 \dots$

(I) Number of spectral lines in Hydrogen atom is

- (a) 8 (b)6 (c) 15 (d) infinity

(II) Which series of hydrogen spectrum corresponding to ultra violet region

- (a) Balmer series (b) Bracket series (c) Paschen series (d)Lymen Series

(III) Which of the following lines of Hydrogen Spectrum belongs to Balmer Series

- (a)1025 $\text{\AA}$ (b) 1218 $\text{\AA}$ (c)4861 $\text{\AA}$ (d)18751 $\text{\AA}$

(IV) Rydberg Constant is

- (a) Universal constant (b) same for same elements
(c) different for different elements (d) none of the above

(V) Hydrogen is excited from ground state to another state with a Principal quantum number equals to 4. Then the number of spectral lines in emission spectra will be

- (a) 3 (b) 5 (c) 6 (d)2

Short Answer type questions (2 marks each)

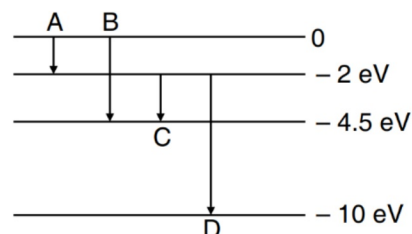
Q11. Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level (ii) the highest permitted energy level to the first permitted level

Q 12. When an electron in hydrogen atom jumps from the third excited state to the ground state, how would the de Broglie wavelength associated with the electron change? Justify your answer.

Q.13 Define the distance of closest approach. An α -particle of kinetic energy 'K' is bombarded on a thin gold foil. The distance of the closest approach is 'r'. What will be the distance of closest approach for an α -particle of double the kinetic energy?

Q.14 In the ground state of hydrogen atom, its Bohr radius is given as 5.3×10^{-11} m. The atom is excited such that the radius becomes 21.2×10^{-11} m. Find (i) the value of the principal quantum number and (ii) the total energy of the atom in this excited state. (2016)

Q.15 The energy levels of a hypothetical atom are given below. Which of the shown transitions will result in the emission of photon of wavelength 275 nm? (2016)



Q.16 Show that the radius of the orbit in hydrogen atom varies as n^2 , where n is the principal quantum number of the atom. (CBSE 2015)

Short answer type questions (3 marks each)

Q 17. A monochromatic radiation of wavelength $975 \text{ \AA}$ excites the hydrogen atom from its ground state to a higher state. How many different spectral lines are possible in the resulting spectrum? Which transition corresponds to the longest wavelength amongst them. (CBSE Sample QP 2018)

Long answer type questions (5 marks each)

Q. 18. (i) Draw a schematic arrangement of the Geiger- Marsden experiment and describe it.
(ii) How did the scattering of α -particle by a thin foil of gold provide an important way to determine an upper limit on the size of the nucleus? Explain briefly.

ANSWERS (SECTION B)

ANSWERS OF MCQ(1mark)

- 1.(a) Thomson's model failed to explain the scattering of alpha particles through large angles in Rutherford experiment.
2. (d) $r=0.53 \text{ }^{\circ}\text{A}$
3. (c) $4 \rightarrow 3$
4. (a) -0.54eV
5. (b) 6
6. (d) angular momentum

Answers of Assertion Reason (1 mark)

7. Correct Answer: B
8. correct Answer B
9. Correct Answer: B

Case Study based question - Answers (5 marks)

10. (I) (d) infinity
- (II) (d) Lyman Series
- (III) (c) $4861 \text{ }^{\circ}\text{A}$
- (IV) (a) Universal constant
- (V) (c) 6

Short answer type questions (2M)- Answers

11. $\Delta E = E_i - E_f$ Ratio=3:4
12. Wave length decreases. $P = \sqrt{2mKE} = h/\lambda$
E is proportional to $1/\lambda$
13. Distance of closest approach: It is the distance of charged particle from the centre of the nucleus, at which the whole of the initial kinetic energy of the (far off) charged particle gets converted into the electric potential energy of the system.

Distance of closest approach (r_c) is given by

$$r_c = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$$

'K' is doubled and r_c becomes $\frac{r}{2}$

14. (i)

$$r = r_0 n^2$$

$$21.2 \times 10^{-11} = 5.3 \times 10^{-11} n^2$$

$$n = 2$$

(ii)

$$\begin{aligned} E &= -\frac{13.6\text{eV}}{n^2} \\ &= -\frac{13.6\text{eV}}{2^2} = -3.4\text{eV} \end{aligned}$$

15. (i)

$$\begin{aligned} \text{Energy of Photon} &= \frac{hc}{\lambda} \\ &= \frac{6.64 \times 10^{-34} \times 3 \times 10^8}{275 \times 10^{-9} \times 1.6 \times 10^{-18}} \text{ eV} \\ &= 4.5 \text{ eV} \end{aligned}$$

(ii) The transition is B.

16.

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2} = \frac{mv^2}{r}$$

$$r = \frac{e^2}{4\pi\epsilon_0 mv^2}$$

$$mv^2 r = \frac{e^2}{4\pi\epsilon_0} \quad \dots\dots\dots(1)$$

According to the Bohr's Postulates,

$$mvr = \frac{nh}{2\pi}$$

$$m^2 v^2 r^2 = \frac{n^2 h^2}{4\pi^2}$$

Using the value of $mv^2 r$ from eqn. (1)

$$\frac{e^2}{4\pi\epsilon_0} mr = \frac{n^2 h^2}{4\pi^2}$$

$$r = \left(\frac{n^2}{m}\right) \left(\frac{h}{2\pi}\right) \frac{4\pi\epsilon_0}{e^2}$$

The above equation show that r is directly proportional to n^2

Short answer type questions (3M)- Answers

17. The corresponding to the given wavelength

$$E = \frac{12400}{\lambda} = \frac{12400}{975} = 12.71 \text{ eV}$$

The excited state

$$E_n - E_1 = 12.71$$

$$-\frac{13.6}{n^2} + 13.6 = 12.71$$

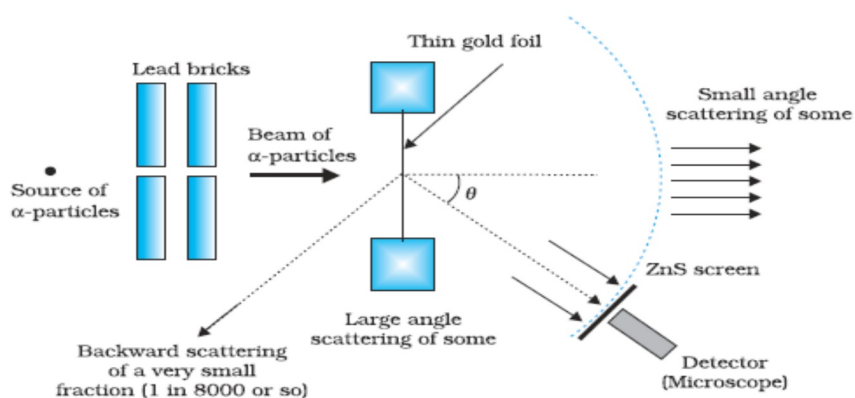
$$\therefore n = 3.9 \approx 4$$

$$\text{total no of spectral lines emitted : } \frac{n(n-1)}{2} = 6$$

Longest wavelength will corresponding to the transition $n=1$ to $n=4$.

Long answer type questions (5M)- Answers

18. (i) Geiger-Marsden experiment: Alpha particles emitted by a $^{214}_{83}\text{Bi}$ radioactive source were collimated into a narrow beam by passing them through lead bricks. The beam was allowed to fall on a thin foil of gold of thickness $2.1 \times 10^{-7} \text{ m}$. The scattered alpha particles were observed through a rotatable detector consisting of zinc sulphide screen and a microscope. The scattered alpha particles on striking the screen produced brief light flashes or scintillations. These flashes may be viewed through a microscope.



From the experiment, it was observed that only a small fraction of number of α -particles rebound back. This shows that the number of α -particles undergoing head on collision is very small. The conclusion is that the entire positive charge of atom is concentrated in a small volume called the nucleus.

(ii) The alpha particles those are incident directly on the gold nucleus, experience a very large force of repulsion and undergo maximum deflection as observed in the experiment. At the distance of head on approach, the entire kinetic energy of α -particle is converted into electrostatic potential energy. This distance of head on approach gives an upper limit of the size of nucleus (denoted by r_0) and is given by

$$E_k = \frac{1}{4\pi\epsilon_0} \cdot \frac{(Ze)(2e)}{r_0}$$

$$r_0 = \frac{1}{4\pi\epsilon_0} \cdot \frac{(Ze)(2e)}{E_k}$$

This is about 10^{-14} m.

SECTION C

Multiple Choice Questions (1 Mark)

Q.1 What is the ratio of radii of orbits corresponding to 1st excited state and ground state of hydrogen atom?

- (a) 2:1 (b) 1:4 (c) 4:1 (d) 1:2

Q.2 The ratio between Bohr radii is

- (a) 1 : 2 : 3 (b) 2 : 4 : 6 (c) 1 : 4 : 9 (d) 1 : 3 : 5

Q.3. An alpha particle of energy 5 Mega eV is scattered through 180 degree by a fixed uranium nucleus. The distance of closest approach is of the order of

- (a) 1 ⁰A (b) 10⁻¹⁰ cm (c) 10⁻¹² cm (d) 10⁻¹⁵ cm

Q.4 The total energy of the electron in the ground state of hydrogen atom is -13.6 eV. The kinetic energy of the electron in the first excited state will be

- (a) 1.7 eV (b) 3.4 eV (c) 6.8 eV (d) 13.6 eV

Q.5 When an electron jumps from 4th orbit to 2nd orbit, one gets the

- (a) second line of paschen series (b) second line of lyman series
(c) second line of balmer series (d) first line of pfund series

Q.6 If P.E and K.E represent potential and kinetic energies of the orbital electron then

- (a) K.E = -P.E/2 (b) K.E = -P.E (c) K.E = -2P.E (d) none of the above

Assertion –Reason Type Questions (1 Mark each)

Answer: A Both are correct and reason is correct explanation of assertion.

Answer: B Both are correct but reason is not the correct explanation of assertion.

Answer: C Reason is wrong.

Answer: D Both are wrong.

Q7. **Assertion:** Electrons in the atom are held due to coulomb forces

Reason: The atom is stable only because the centripetal force due to Coulomb's law is balanced by the centrifugal force.

Q8. **Assertion:** Rydberg's constant varies with the mass number of given element.

Reason: The reduced mass of electron is dependent on the mass of nucleus only.

Q9. Assertion: In Lyman series, the ratio of minimum and maximum wavelength is $\frac{3}{4}$

Reason: Lyman series constitute spectral lines corresponding to transition from higher energy to ground state of hydrogen atom.

Case study based question (5 marks)

Q10. ENERGY OF ELECTRON FOR HYDROGEN ATOM

Electrons are revolving around the nucleus in particular stable orbits. The energy of the electron is increasing as we go from the orbit closer to nucleus to outer side. The ground state energy is the lowest energy and it is -13.6 eV for hydrogen atom. Thus, the minimum amount of energy required to remove or free the electron from the ground state is the ionisation energy and it has value +13.6 eV. When electrons jumps from higher energy orbit to lower energy orbit emits energy in the form of photons which are in the form of spectral lines and called as emission lines. The light emitted by the ordinary source of light consist of different wavelength. But the laser light is the monochromatic one which emits light of single wavelength. In case of hydrogen atom, the ground state energy is that energy state for which $n = 1$. And the states for which $n > 1$, all are the excited states. Where n shows the principal quantum number.

(I) In hydrogen atom, the energy corresponding to principal quantum number $n = 2$ is ____

- (a) -13.6 eV (b) -3.4 eV (c) +13.6 eV (d) +3.4 eV

(II) For ground state of hydrogen atom the value of principal quantum number is ____

- (a) $n = 2$ (b) $n = 0$ (c) $n = 1$ (d) $n = \text{infinity}$

(III) The minimum energy required to remove the electron from the ground state of the hydrogen atom is called as ____

- (a) excitation energy (b) ionisation energy
(c) ground state energy (d) excited state energy

(IV) The acronym LASER stands for?

(V) If $n = \text{infinity}$ then what is the energy of the state and what does it means?

Short answer type questions (2 marks each)

Q 11. What is the maximum number of spectral lines emitted by a hydrogen atom when it is in the third excited state? Which one will have lowest wavelength?

Q 12. The wavelength of the second line of Balmer series in Hydrogen atom is 4861 Angstrom. Calculate the wave length of the first line.

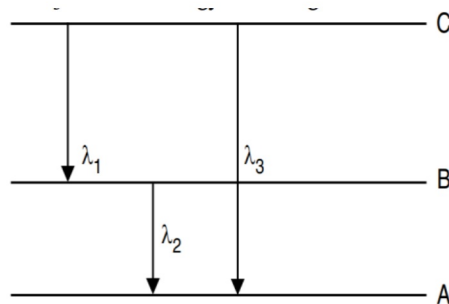
Q13. Calculate the shortest wavelength in the Balmer series of hydrogen atom. In which region (infra-red, visible, ultraviolet) of hydrogen spectrum does this wavelength lie?

Q14 The ground state energy of hydrogen atom is -13.6 eV . If an electron makes a transition from an energy level -1.51 eV to -3.4 eV , calculate the wavelength of the spectral line emitted and the series of hydrogen spectrum to which it belongs. (CBSE 2017)

Q15 Energy of electron in first excited state in Hydrogen atom is -3.4 eV . Find KE and PE of electron in the ground state. (CBSE SQP 2019-20)

Short answer type questions (3 marks each)

Q16. (i) State Bohr's quantization condition for defining stationary orbits. How does de-Broglie hypothesis explain the stationary orbits? (ii) Find the relation between the three wavelengths λ_1 , λ_2 and λ_3 from the energy level diagram shown below. (2016)



Long answer type questions (5 marks each)

Q.17 (i) Write two important limitations of Rutherford model which could not explain the observed features of atomic spectra. How were these explained in Bohr's model of hydrogen atom ? Use the Rydberg formula to calculate the wavelength of the H_α line. (Take $R = 1.1 \times 10^7 \text{ m}^{-1}$).

(CBSE 2015)

(ii) Using Bohr's postulates, obtain the expression for the radius of the n^{th} orbit in hydrogen atom.

ANSWERS (SECTION C)

MCQs- Answers (1 mark)

1.(c)Ground state, $n=1$

Excited state, $n=2$

r proportional to n^2

$$r_2/r_1 = 2^2/1^2$$

$$r_2/r_1 = 4 = 4:1$$

2. (c) 1:4:9

3. (c) 10^{-12} cm

4. (b) 3.4 eV

5. (c)second line of balmer series

6. (a) $K.E = -P.E/2$

Assertion Reason - Answers (1 mark)

7. Correct Answer: C

8. Correct Answer: B

9. Correct Answer: B

Case Study based question – Answers (5 marks)

10. (I) b) -3.4 eV

(II) c) $n = 1$

(III) b) ionisation energy

(IV) The acronym LASER stands for Light Amplification by Stimulated Emission of Radiation.

(V) When the principal quantum number $n = \text{infinity}$ then the corresponding state is having energy 0 eV. And this energy of atom is possible only when electron is totally removed from the nucleus and hence it goes to rest.

Short answer type questions (2M) – Answers

11. third excited state, $n_2 = 4$, and $n_1 = 3, 2, 1$ Hence there are 3 spectral lines.

Greater the energy lower the wavelength

12. For the first line in balmer series

$$\lambda_1 = R(1/2^2 - 1/3^2) = 365R$$

For second balmer line:

$$4861 = R(1/2^2 - 1/4^2) = 163R \text{ Divide both equations:}$$

$$\lambda = 4861 \times 27/20 = 6562.35 \text{ } ^0\text{A}$$

13. For shortest wavelength in the Balmer series: $n_1=2$ $n_2=\infty$

$$\lambda_{\min} = 3.646 \times 10^{-7} \text{ m} = 364.6 \text{ nm. This wavelength lies in the ultraviolet region}$$

$$14. \quad h\nu = \frac{hc}{\lambda} = (E_2 - E_1)$$

$$\lambda = \frac{hc}{(E_2 - E_1)}$$

$$\lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{[-0.85 - (-3.4)] \times 6 \times 10^{-19}}$$

$$\lambda = 4.87 \times 10^{-7} \text{ m}$$

This wavelength belongs to Balmer series.

15. Energy of electron in $n=2$ is 3.4eV

Energy ground state = -13.6eV

$$\text{K.E.} = -\text{T.E.} = +13.6 \text{ eV}$$

$$E_n = \frac{x}{n^2}$$

$$= -13.6 \text{ eV} = \frac{x}{2^2}$$

Energy of ground state $x = -13.6 \text{ eV}$.

$$\text{P.E.} = 2\text{T.E.} = -2 \times 13.6 \text{ eV} = -27.2 \text{ eV}$$

Short answer type questions (3M) – Answers

16. (i) $L = \frac{nh}{2\pi}$ i.e. angular momentum and orbiting electron is quantized

According to de-Broglie hypothesis

$$\text{linear momentum, } p = \frac{h}{\lambda}$$

for circular orbit, $L = r_n p$ where r_n is radius of n^{th} orbit

$$= \frac{r_n h}{\lambda}$$

$$\text{also } L = \frac{nh}{2\pi}$$

$$\frac{r_n}{\lambda} = \frac{nh}{2\pi}$$

$$2\pi r_n = n\lambda$$

∴ circumference of permitted orbits are integral multiples of the wavelength λ

(ii)

$$E_C - E_B = \frac{hc}{\lambda_1} \quad \dots\dots(1)$$

$$E_B - E_A = \frac{hc}{\lambda_2} \quad \dots\dots(2)$$

$$E_C - E_A = \frac{hc}{\lambda_3} \quad \dots\dots(3)$$

According to equations (1) and (2)

$$E_C - E_A = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} \quad \dots\dots(4)$$

Using equation (3) and (4), we get

$$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$$\frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$

Long answer type questions (5M) – Answers

17. (i) Limitations of Rutherford's atomic model are as follows:

(1) According to Rutherford's atomic model, the electron orbiting around the nucleus continuously radiates energy because of its acceleration, due to which the atom will not remain stable. So, Rutherford could not explain the stability of an atom.

(2) Since electron spirals inwards, its angular velocity and frequency change continuously, therefore it will emit a continuous spectrum. Therefore, this model could not explain the line spectra of hydrogen.

According to Bohr's model of hydrogen atom, Electron in an atom can revolve in certain stable orbits without emitting the energy. Energy is released or absorbed only when an electron jumps from one stable orbit to another stable orbit resulting in a discrete spectrum.

$$\frac{1}{\lambda} = R \left(\frac{1}{2^1} - \frac{1}{3^2} \right)$$

$$\frac{1}{\lambda} = 1.1 \times 10^7 \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$\lambda = 656.3 \text{ nm}$$

(ii) **From Bohr's first postulate,**

coloumb force F = centripetal force F_{cp}

$$\therefore \frac{Ze^2}{4\pi\epsilon_0 r_n^2} = \frac{m_e v_n^2}{r_n}$$

$$v_n^2 = \frac{Ze^2}{4\pi\epsilon_0 r_n m_e} \dots\dots\dots(1)$$

according to Bohr's second postulate,

$$m_e r_n v_n = \frac{nh}{2\pi}$$

$$m_e^2 r_n^2 v_n^2 = \frac{n^2 h^2}{4\pi^2}$$

$$v_n^2 = \frac{n^2 h^2}{4\pi^2 m_e^2 r_n^2} \dots\dots\dots(2)$$

iv from equation (1) and (2)

$$\frac{n^2 h^2}{4\pi^2 m_e^2 r_n^2} = \frac{Ze^2}{4\pi\epsilon_0 r_n m_e}$$

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m_e Z e^2}$$

$$r_n = \left(\frac{\epsilon_0 h^2}{\pi m_e^2 Z e^2} n^2 \right)$$

this is the required equation for radius of n^{th} orbit.
